

Measuring Watermark–Detector Interference: Controls, Score Attenuation, and Checkpoint Dependence

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Abstract—Prior work reports that embedding a provenance watermark can *harm* a passive synthetic-image detector inspecting the same pixels. That claim has a confound: any imperceptible perturbation shifts a detector trained on clean images, so degradation after marking need not be watermark-specific. We introduce a control the prior literature does not apply — a payload-free perturbation matched per image to the watermark’s own PSNR — and re-measure the effect on 20,000 held-out images under hypotheses fixed before the data was scored.

The control reverses the expected conclusion. On an EfficientNet-B6 detector, the degradation attributed to watermarking belongs to the control: matched-PSNR noise attenuates the detector’s log-odds evidence toward zero by $\approx 16\%$ (slope 0.83–0.85), while watermarking leaves it essentially intact (slope 0.97–0.99). We then test whether this attenuation is a property of detection or of one checkpoint, by repeating the measurement on a second, provenance-disclosed detector matched to the first on baseline AUC on the same 8,000 images. The asymmetry does not replicate and its sign reverses, closing off a weak-detector explanation: the effect is checkpoint-dependent, not a general property of passive detection.

What generalises past this dataset is structural: attenuation is a monotone rescaling and is exactly invisible to AUC while it still degrades calibration; a mean-shift statistic cannot distinguish attenuation from translation, but a slope estimate can. In support: the tested robust watermark’s payload survives a face-replacement-scale splice at bit error rate 0.0000, and a detector-floor precondition found five of six public checkpoints operating at chance, one advertising 98.70% accuracy at AUC = 0.54 in use.

Index Terms—digital watermarking, deepfake detection, image forensics, provenance, score calibration, AUC, checkpoint generalisation, adversarial robustness, pre-registration

I. INTRODUCTION

Two defences against synthetic media are converging in deployment. Proactive provenance marking embeds a signal at creation time so origin can be verified later; passive detection infers manipulation from residual synthesis artefacts. Both are computed from the same pixels, so they are not independent: Wu et al. [1] report that watermarking models built for genuine images can *harm* a deployed detector, because watermark signals overlap the forgery signals detection relies on; the same pathology is reported independently in audio [2]. If that holds generally, a detector’s score distribution shifting on marked media invalidates any decision threshold calibrated on unmarked media.

That argument has a load-bearing confound: a detector trained on clean images responds to *any* imperceptible perturbation, so degradation after marking need not implicate the watermark’s structure. No prior work in this space separates the two. We do, with a payload-free perturbation matched per image to the watermark’s own PSNR, reporting every interference quantity net of that arm, with a detector-floor precondition and a gate threshold fixed before scoring.

Our pre-registered gate returned a null on ranking. Auditing why turned the result over: the watermark contributed almost nothing to the quantity we had also examined, and the *control* carried the effect — not as a shift, but as an attenuation of the detector’s evidence toward zero. Attenuation with near-zero intercept is monotone and therefore invisible to AUC by construction, which is why a ranking gate returned a null while a real effect was present.

That corrected mechanism then failed its own replication. Every number above came from one third-party checkpoint with undisclosed training data, which makes “attenuation is a property of detection” and “attenuation is a property of this model” observationally identical. We built a second detector with disclosed provenance — a frozen ImageNet ResNet-18 with a logistic head fitted only on the training split — and repeated the measurement. The asymmetry did not appear (\$V-C): the payload-free arm attenuated *less* than the watermarked one, on images where the two detectors are matched to within 0.003 AUC. Sensitivity to matched-PSNR noise is therefore a fact about a checkpoint, not about passive detection, and the field’s usual apparatus — one detector, no control, AUC as the outcome — cannot see either confound.

This paper’s spine is four results in sequence: a control that overturns an attribution, the structural reason the field’s standard metric misses it, a replication that bounds how far the mechanism generalises, and what survives that bounding. Three supporting measurements are reported as such, not as independent claims of equal weight.

- 1) **A control that changes the sign of the conclusion.** Every interference quantity is reported net of a payload-free perturbation matched per image to the watermark’s own PSNR. Without it, our own headline number would have read as watermark-induced degradation that in fact belongs to the control.
- 2) **The right estimand is a slope.** A mean-shift statistic cannot distinguish attenuation from translation; attenuation with near-zero intercept is *exactly* invisible to AUC

because it is monotone (verified by unit test), which is why a ranking gate can return a null while a real effect is present.

- 3) **A replication failure that bounds the claim (§V-C).** On EfficientNet-B6, matched-PSNR noise attenuates evidence by $\approx 16\%$ while watermarking does not; on a second, provenance-disclosed detector matched to within 0.003 AUC on the same images, the asymmetry vanishes and reverses. Two detectors establish checkpoint-dependence, not which population of detectors exhibits the effect, and we claim no more.
- 4) **Supporting measurements.** The tested robust watermark survives a face-replacement-scale splice at BER 0.0000 (§V-D); attenuation degrades a well-specified calibrator’s ECE $4\text{--}5\times$ but does not reliably move decisions (§V-E); and a floor check caught five of six public checkpoints at chance, one advertising 98.70% accuracy.

II. RELATED WORK

Proactive watermarking. A substantial line of work embeds a mark at creation time to establish provenance, increasingly unifying provenance with detection or localisation [3], [4]. Our concern is orthogonal: what embedding a mark does to a passive detector inspecting the same pixels, treating watermark strength (one hand-crafted, one learned scheme) as an experimental axis rather than proposing a new one.

A larger line of work responds to the failure mode we measure in §V-D: robust, spatially redundant watermarks survive localised tampering for the same reason they survive benign processing, which makes them poor at answering “was this edited”. Semi-fragile and separable-decoder schemes break that coupling — SepMark’s two decoders (robust Tracer, fragile Detector) [5], fractal- and landmark-derived constructions [6], [7], image-bit-steganography reframings [8], [9], and wavelet/graph, diffusion-conditioned, and GIF-specific embeddings [10], [11], [12] all exist because the failure mode is understood well enough to design around. A second, orthogonal failure surface is diffusion-based regeneration, which degrades a payload through global stochastic transformation rather than local editing [13], [14], [15]. Two different mechanisms — regeneration and splicing — converge on the same conclusion about robust watermarking’s limits.

Passive detection. Reported accuracy transfers poorly once inputs are perturbed [16]; watermark robustness has been benchmarked in the same spirit [17]. Our E0 result is a sharp instance: a detector advertising 98.70% accuracy operates at chance on a distribution it was not tuned for.

Interference. The premise this work tests is Wu et al.’s [1] argument that watermarks devised for genuine images “may harm the deployed Deepfake detectors ... since the watermarks are prone to overlap with the forgery signals used for detection”, answered with AdvMark, fine-tuning the watermark itself. The same pathology is reported independently in audio [2]. Guo et al. [18] benchmark passive against watermark-based detection head-to-head but never combine them, so whether marking degrades the passive detectors

being benchmarked is not asked. Two gaps follow, which this paper addresses: no prior interference result controls for generic perturbation sensitivity, and none is tested under a pre-registered null. Consequences for a fused decision rule remain open; our corpus cannot settle that (Limitations).

III. METHOD

A. Provenance evidence as a likelihood ratio

Let the recovered payload contain k_e bit errors out of L . Under intact provenance, errors are channel noise, $k_e \sim \text{Bin}(L, p_0)$; under no valid provenance the recovered bits are uninformative, $k_e \sim \text{Bin}(L, 1/2)$. The provenance log-likelihood ratio is

$$z_P = L \log 2 + k_e \log p_0 + (L - k_e) \log(1 - p_0), \quad (1)$$

with p_0 estimated on a held-out clean-transform calibration set. z_P is in log-odds units and is defined only when a mark is present.

B. Interference and the evidence-transfer estimator

For watermark scheme s and class y ,

$$\Delta_\mu(s, y) = \mathbb{E}[V \mid W=1, S=s, Y=y] - \mathbb{E}[V \mid W=0, Y=y], \quad (2)$$

$$\Delta_{\text{AUC}}(s) = \text{AUC}(V; W=1, S=s) - \text{AUC}(V; W=0). \quad (3)$$

The attenuation analyses of §V-B–V-C add one estimator: for each arm s we fit $v_s = a_s + b_s v_{\text{clean}}$ by OLS, *per arm and per detector separately*, since the question is whether an arm’s slope differs from 1, not whether arms differ from each other by assumption. Every fit is *paired* on the same image. Confidence intervals on b_s and on slope differences are 95% percentile bootstrap over images (resampling an image moves all its arms together), 800–2,000 resamples depending on the analysis. Where two detectors’ slopes are compared (§V-C), both fits use the identical image set, so a difference in b_s cannot be an artefact of a difference in sample.

C. Fusion and abstention

$$\hat{y} = \sigma(\beta_0 + \beta_1 z_P + \beta_2 V + \beta_3 W + \beta_4 (W \cdot V)). \quad (4)$$

The independence assumption implicit in naive fusion forces $\beta_4 = 0$; $\beta_4 \neq 0$ measures interference inside the decision rule. With costs c_{FN}, c_{FP}, c_R , Chow’s rule [19] gives

$$\tau_{lo} = \frac{c_R}{c_{FN}}, \quad \tau_{hi} = 1 - \frac{c_R}{c_{FP}}, \quad (5)$$

valid iff $c_R/c_{FN} + c_R/c_{FP} < 1$, and optimal only if $\hat{y}$ is calibrated — the dependency this paper is about.

IV. EXPERIMENTAL SETUP

Data. A public corpus of StyleGAN-generated faces against real photographs, balanced by class: whole-image synthesis, *not* face-swap or reenactment, hence “synthetic-image detection” throughout (Limitations states the cost of that substitution). The exploratory pass used 150+150 images; the confirmatory analysis scores 20,000 held-out images across five arms (clean, two schemes, two matched nulls), and §V-C adds a second detector over 8,000 of them — 140,000 detector evaluations total, released as raw per-image scores. Splits are stratified at the image level (no identity metadata), so confidence intervals are sample-level bootstraps.

Watermarks. Two schemes spanning the hand-crafted/learned divide, pretrained: DWT-DCT-SVD (transform-domain) and RivaGAN (learned encoder/decoder), both at a 32-bit payload. Measured imperceptibility: 36.9 dB / 0.967 SSIM and 40.2 dB / 0.983 respectively. For every image and scheme we additionally generate a payload-free perturbation binary-searched to match *that image’s* PSNR to within 0.5 dB; all interference quantities are net of this arm. PSNR matching is not spectral matching, so the control bounds generic perturbation sensitivity without fully isolating watermark structure.

Detectors. Six public checkpoints were screened; one, an EfficientNet-B6, clears the floor of §V-A, used zero-shot with raw logit difference $V = \ell_{\text{fake}} - \ell_{\text{real}}$ rather than softmax confidence. Its training data is undisclosed, so results are within-model, within-image deltas. Because one undisclosed checkpoint cannot separate a property of detection from a property of a model, we add a second detector we train ourselves: a frozen ImageNet ResNet-18 feature extractor (512-d) with an ℓ_2 -regularised logistic head fitted on the training split alone. Validation and test are untouched by construction, and the head is deliberately linear — the most architecturally distant thing that still clears the floor, so a shared effect is evidence of generality and a divergent one is evidence against it.

Protocol. The detector floor, the gate threshold on $\Delta\text{AUC}_{\text{net}}$, and the fallback framing for a null were fixed before scoring. Every number below is produced by released code from a single configuration file.

V. RESULTS

A. Detector floor check (E0)

A detector near chance makes ΔAUC a measurement of noise. We screened six public checkpoints; **five operate at chance**:

Two checkpoints agree to four decimal places across 300 images, suggesting one is a re-upload of the other, so the pool of genuinely independent public detectors is smaller than a hub search implies. The gap between a card’s 98.70% and $\text{AUC} = 0.54$ in use is not a rounding error; a floor check is what stands between such a checkpoint and a table of confident noise. It also bounds this study — only one public detector clears the floor — so rather than state that as a limitation we removed it by training a second, disclosed detector (§V-C).

TABLE I
BASELINE AUC OF EVERY SCREENED CHECKPOINT.

Checkpoint	Baseline AUC	Status
EfficientNet-B6 (Skullyy)	0.8952	clears floor
ViT (Wvolf)	0.5365	chance; card claims 98.70%
DaMsTaR/Detecto	0.5365	chance
prithivMLmods/v2	0.5299	chance
dima806	0.5296	chance
Hemg	0.4576	below chance

TABLE II
CONFIRMATORY PRE-REGISTERED TESTS, NET OF THE CONTROL ARM.

Scheme	$\Delta\mu_{\text{net}}$ (H1)	$\Delta\text{AUC}_{\text{net}}$ (H2)
DWT-DCT-SVD	+0.837 [+0.807, +0.867]	+0.0049 [+0.0032, +0.0066]
RivaGAN	+1.353 [+1.317, +1.387]	−0.0051 [−0.0071, −0.0033]

B. Confirmatory result ($n = 20,000$)

Chronology, as recorded in a git-timestamped pre-registration: (1) an exploratory pass at $n=300$ on *test* found no effect on ranking; (2) repairing an unrelated defect exposed $\Delta\mu_{\text{net}}$, previously unexamined, on a 36-image slice — it was large; (3) because that observation was post-hoc, we wrote a second pre-registration committing H1/H2/H3 and their analysis code before scoring further; (4) confirmatory hypotheses were tested on *validation* ($n=20,000$), untouched by (1)–(2). One disclosed deviation: at 2% of the run, a pipeline-verification script incidentally exposed interim values consistent with H1/H2; the run was not altered in response, and the full log is released with the code.

Both tests pass, but $\Delta\mu_{\text{net}}$ **is the wrong estimand**: an audit shows the watermark contributes almost nothing for DWT-DCT-SVD (−0.009 watermark–clean vs. −0.846 null–clean; net +0.837 is the control falling). Nor is it a location shift — it ranges −0.36 to +2.28 across clean-score quintiles. Regressing each arm on the clean score:

The mechanism is attenuation: random perturbation shrinks this detector’s log-odds evidence toward zero by $\approx 16\%$ ($b \approx 0.84$, $a \approx 0$) while watermarking leaves it intact ($b \approx 0.97$ –0.99). §V-C shows this is checkpoint-specific; the consequences below are not, since they are properties of the statistics. First, **attenuation is invisible to AUC by construction** (monotone rescaling, verified by unit test), which is why our AUC gate returned a null while a real effect was present. Second, **attenuation is precisely what breaks calibration**, shrinking log-odds toward zero and changing the score-to-posterior mapping. Effect sizes (clean-score sd 7.04): net effects of 0.12–0.19 sd, paired Cohen’s d_z of 0.38/0.55 — moderate, and class-dependent: the null arm erodes evidence on manipulated images roughly $3\times$ more than on real ones.

C. Replication on a provenance-disclosed detector

One undisclosed checkpoint cannot separate “attenuation is a property of detection” from “attenuation is a property of this model”. We built a detector with disclosed provenance — frozen ImageNet ResNet-18, logistic head on *training* alone

TABLE III
EVIDENCE-TRANSFER REGRESSION, EFFICIENTNET-B6.

Arm	slope b	intercept a
DWT-DCT-SVD	0.968	+0.125
null (PSNR-matched)	0.834	−0.158
RivaGAN	0.990	+0.649
null (PSNR-matched)	0.850	−0.125

TABLE IV
CROSS-DETECTOR SLOPE COMPARISON, IDENTICAL IMAGES,
800-RESAMPLE BOOTSTRAP.

Detector	Scheme	wm b [95% CI]	null b [95% CI]	null−wm
EffNet-B6	DWT	0.966 [.962, .971]	0.832 [.825, .839]	−.134
EffNet-B6	Riva	0.987 [.983, .992]	0.848 [.841, .856]	−.139
ResNet-18	DWT	0.956 [.951, .962]	0.971 [.967, .974]	+0.014
ResNet-18	Riva	0.949 [.943, .954]	0.976 [.972, .979]	+0.027

— and refit the regression on the same 8,000 images restricted to the identical set both detectors scored:

The asymmetry does not replicate, and its sign reverses: on the disclosed detector the null attenuates *less* than the watermark, against 16% more on EfficientNet-B6, with tight, disjoint intervals.

The escape that a linear probe is too weak to show the effect is closed by matched baseline AUC on the same images: 0.8703 (EfficientNet-B6) vs. 0.8725 (ResNet-18 probe), moving near-identically under every arm. What differs is how the score *distribution* moves — exactly what the regression measures and AUC cannot see. We therefore withdraw “imperceptible noise attenuates evidence while watermarking does not” as a general claim: EfficientNet-B6’s sensitivity is a fact about that model (plausibly training augmentation, or its 528×528 processing resolution vs. the probe’s 224×224). The control arm is only more strongly motivated by this: without it, a study would report the null’s -0.846 shift as the watermark’s; with it but one detector — every interference paper we are aware of, and this one until this section — it would report a checkpoint artefact as a property of detection. The structural results ($\Delta\mu$ conflates attenuation with translation; AUC is exactly blind to it) are analytic and unaffected by replication.

D. Robust watermarks cannot see localised manipulation

The source framework routes “is the watermark present, and has it been altered?” into detection. We supplied the missing manipulation step — watermarking an authentic image, *then* splicing a donor region in with Poisson blending — and measured BER against manipulated area:

At 10–25% coverage — what a face replacement costs — the payload survives *intact* (BER 0.0000); obliterating 95% still leaves BER at 0.26, far from the destruction threshold of 0.5. This is a property of robust watermarking, not our corpus: the redundancy that survives JPEG, resizing and re-encoding is what survives a region being replaced. We do not claim this beyond the scheme and mechanism tested, though it is generic enough that a robust, spatially redundant scheme behaving otherwise would surprise us. Consequently, a

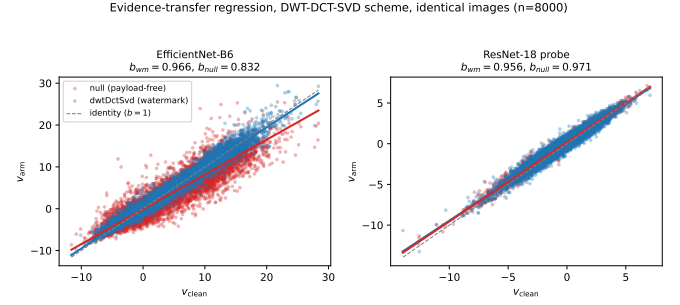


Fig. 1. Evidence-transfer regression on the same 8,000 images, DWT-DCT-SVD scheme. EfficientNet-B6 (left) shows the null arm (red) pulled below identity while the watermark (blue) tracks it; the ResNet-18 probe (right) shows the opposite ordering.

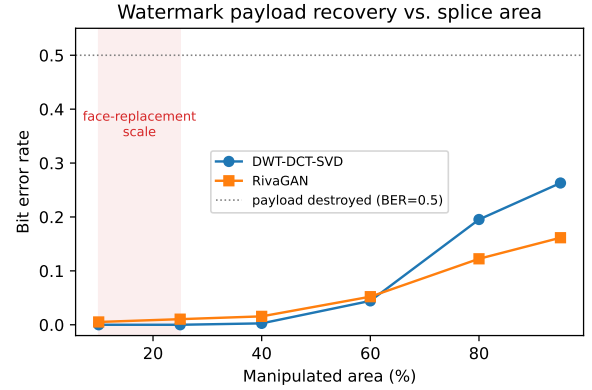


Fig. 2. BER of the recovered payload vs. spliced area. The shaded band is realistic face-replacement coverage; both curves are near-zero across it.

fused decision rule over this channel combines one informative signal with a constant — the planned fusion comparison is ill-posed here, not merely underpowered, which is why §II’s semi-fragile schemes exist as a response to exactly this failure. The crossover sits at 40–60% manipulated area, far outside realistic face manipulation.

E. Calibration under attenuation

Attenuation degrades calibration, not just rank; we measure this with ECE under Platt and isotonic calibration [20], fit on the *clean* half of a calibration split and frozen before evaluation:

Isotonic shows attenuation degrading ECE $4\text{--}5\times$ over clean while watermarking degrades it far less; Platt shows the null arms *improving*, which we resolve rather than cherry-pick: Platt’s clean-arm ECE is already $13\times$ isotonic’s (raw logits span -11.6 to $+36.8$, $\text{sd}=7.04$), so its baseline is broken before any perturbation, and attenuation happens to partially cancel that misspecification. The claim survives for calibration *quality* and fails for decision *risk*: isotonic DRD is 0.0055 (clean), 0.0145/0.0054 (nulls) — small and inconsistent in direction. Attenuation reaches the probability estimates without reliably moving decisions at these costs.

TABLE V
EXPECTED CALIBRATION ERROR BY ARM AND CALIBRATOR.

Arm	Platt ECE	Isotonic ECE	slope vs. clean
clean	0.1273	0.0094	1.000
DWT-DCT-SVD	0.1228	0.0154	0.968
RivaGAN	0.1353	0.0290	0.990
null[DWT]	0.0958	0.0492	0.834
null[RivaGAN]	0.1009	0.0456	0.850

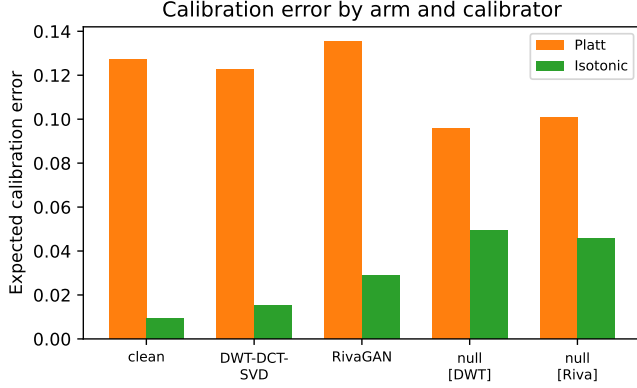


Fig. 3. ECE by arm under each calibrator. Isotonic shows attenuation degraded calibration; Platt shows the opposite, resolved by its badly misspecified clean-arm baseline (0.1273, 13 $\times$ isotonic’s).

F. Why the fusion comparison is not reported, and transform robustness

The planned fusion comparison (F0–F5, with/without an interaction term) is not reported: §V-D shows the provenance channel constant under manipulation, so any recovered β_4 would describe sampling noise, not signal. We instead report a different fusion question — whether β_4 responds to a standard transform on otherwise unmanipulated images — at reduced scale, flagged as a first look: $n=400$ (120/280 split, held-out *test*, disjoint from validation), EfficientNet-B6, seven transforms, *no bootstrap interval computed*.

F0 and F4 AUC move together under compression/resize — the detector’s own fragility, not a watermark effect. β_4 stays small (< 0.08) for six of seven transforms; resize-to-0.5 $\times$ is an outlier ($\beta_4 = +0.221$, DRD 0.050). At $n=280$ with no interval, this is flagged for a CI-backed rerun, not claimed. The ρ sweep (fusion fit once on calibration, evaluated on watermarked-fraction mixtures) found AUC/ECE/DRD essentially flat across $\rho \in \{0.05, 0.25, 0.5, 1.0\}$ for both schemes:

— consistent with the provenance channel’s weak contribution throughout: if it does not move the outcome at $\rho=1$, its prevalence will not matter either.

VI. LIMITATIONS

Provenance channel inertness, measured not assumed.

We supplied the manipulation step the source framework presupposes and measured the channel still inert (§V-D) — a finding, not a corpus artefact. Our splice is faithful to the provenance mechanics but not to a learned face swap’s

TABLE VI
TRANSFORM ROBUSTNESS, F0 (NO INTERACTION) VS. F4 (WITH IT), $N=400$.

Transform	F0 AUC	F0 DRD	F4 AUC	F4 DRD	F4 β_4
JPEG(90)	0.899	0.006	0.897	0.008	+0.004
JPEG(70)	0.776	0.003	0.819	0.005	−0.022
JPEG(50)	0.721	0.001	0.794	0.005	+0.038
Resize(0.75)	0.743	0.003	0.825	0.008	+0.048
Resize(0.5)	0.710	0.001	0.825	0.050	+0.221
Bright(+20%)	0.909	0.034	0.909	0.031	+0.056
Bright(−20%)	0.841	0.003	0.848	0.006	−0.072

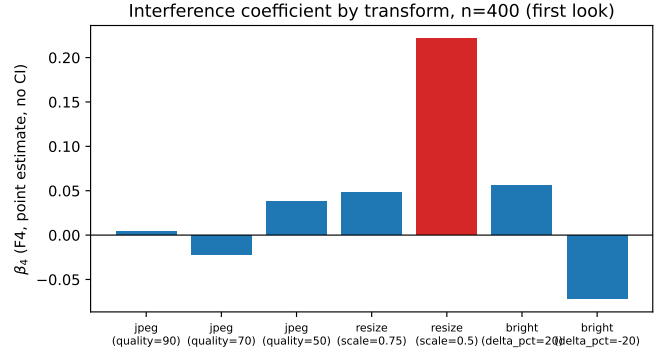


Fig. 4. β_4 per transform (F4, $n=280$, no CI). Resize-to-0.5 $\times$ (shaded) is 3–6 $\times$ every other condition and is flagged, not claimed.

detector-facing realism, so claims concern localised tampering generally, not modern face-swap detectability specifically.

Statistical power. The $n=300$ exploratory pass has AUC standard error ≈ 0.03 , larger than the effects measured, so its null means “undetectable at this size”. Confirmatory and replication analyses ($n=20,000/8,000$) do not have this problem; reported intervals, not point estimates, bound excluded effect sizes.

Synthesis type. Whole-image GAN synthesis vs. real photographs, not face-swap/reenactment as in standard benchmarks [21]; detectors keying on generator fingerprints need not generalise to different manipulation traces.

Detector provenance and count. The strong detector’s training data is undisclosed (mitigated by within-model, within-image deltas); the disclosed detector is a linear probe on generic features, not an engineered detector. Two detectors establish non-universality, not which population of detectors exhibits the effect — that needs a panel, and §V-A shows floor-clearing detectors are the scarce resource.

Other. Interference is measured zero-shot; a task-trained detector is untested. Decision thresholds use an assumed cost triple and swept, not measured, ρ .

VII. DISCUSSION

What a null does and does not mean. We did not detect [1]’s effect in our setting, which is a boundary condition, not a refutation: their detectors are trained for the manipulation family they evaluate and their images are manipulated *after* marking, so watermark and forgery signals coexist as their argument describes. Ours are zero-shot detectors on whole-image synthesis marked after the fact.

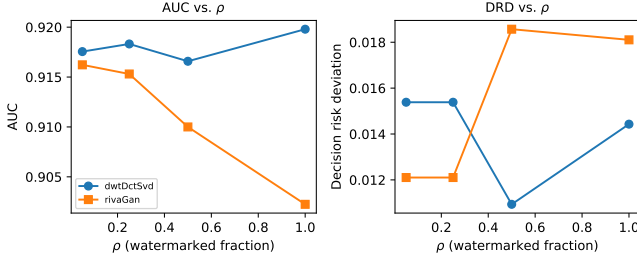


Fig. 5. AUC and decision-risk deviation vs. watermarked fraction ρ , model F4. Both are flat, consistent with the provenance channel’s weak contribution elsewhere in this paper.

Ranking vs. calibration. ΔAUC is silent on translation, which leaves ranking untouched while invalidating a posterior calibrated on unmarked media — exactly our own pre-registered gate’s asymmetry ($\Delta\text{AUC}_{\text{net}}$ tested; a calibration-centred thesis needs $\Delta\mu_{\text{net}}$). Any reading of the latter here is post-hoc; the correct response is a fresh pre-registration, not a reinterpretation.

One checkpoint is not a detector. §V-C’s replication failure is the result we would have preferred to avoid and the one most worth reporting: a 16% attenuation, sign-reversed on a model matched to 0.003 AUC. The claim is not that our probe is the right second detector — it is a weak one — but that a difference on one checkpoint is not yet a finding, and the cheapest fix (a frozen backbone with a linear head) is within any paper’s budget.

Detector floors deserve to be standard. A precondition check cost almost nothing and caught a checkpoint at chance while advertising 98.70%. Any study measuring a difference in detector behaviour inherits the assumption that the detector behaves at all.

VIII. CONCLUSION

We set out to measure whether provenance watermarking interferes with passive synthetic-image detection; the defensible answer is narrower than the one we started toward. Against a PSNR-matched control on 20,000 images, an EfficientNet-B6 detector’s evidence is attenuated $\approx 16\%$ by imperceptible noise and left intact by watermarking — and against a second, disclosed detector matched to 0.003 AUC, that asymmetry reverses. Two detectors show the effect is checkpoint-dependent, not which population it belongs to; we claim only the former, and only because we ran both the control arm and the second model, neither standard practice here.

What generalises is structural: a mean-shift statistic conflates attenuation with translation, and AUC is exactly blind to monotone attenuation, so gating on ΔAUC while measuring with $\Delta\mu$ can be simultaneously wrong about the mechanism and unable to see it. The tested robust watermark carries no information about the manipulation it is meant to flag (BER 0.0000 at realistic coverage), so fusion over it combines one source with a constant. A floor check caught five of six public detectors at chance, one advertising 98.70% accuracy — a precondition for trusting the above.

The useful reading is not that interference is absent, but that the apparatus commonly used to look for it — one checkpoint, no control, AUC as the outcome — cannot tell an effect from its control, a model from its architecture family, or a calibration-relevant change from a null.

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